Qinyuan Wu

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Research Interests

My research focuses on how AI models organize internal knowledge and under what conditions they can reliably execute actions within larger computing systems. By contrasting descriptive model behavior against normative frameworks, I identify critical reliability gaps to help build more trustworthy, human-centered AI systems.

Education

CS@Max Planck & Saarland University Oct 2022 – Present
Ph.D. in Computer Science
Advisors: Prof. Krishna P. Gummadi (MPI-SWS) & Prof. Muhammad Bilal Zafar (Ruhr-Universität Bochum)

University of Electronic Science and Technology of China (UESTC) Sep 2018 – Jun 2022
B.Sc. in Mathematics-Physics Fundamental Science
Yingcai Honors College (top 2% selected)

Publications by Research Theme

★ First or equal-first author. Papers are grouped by research question, with lead-author work listed first.

Research thread: How experience becomes knowledge → how knowledge is estimated and recalled → how knowledge and external evidence guide reliable action.

1. Learning & Encoding

How do memorization and learning interact, and when does stored knowledge support generalization?

- ★ **Rote Learning Considered Useful: Generalizing over Memorized Data in LLMs.**
  [ICLR 2026] *The Fourteenth International Conference on Learning Representations*
  Qinyuan Wu, Soumi Das, Mahsa Amani, Bishwamittra Ghosh, Mohammad Aflah Khan, Krishna P. Gummadi, Muhammad Bilal Zafar.

- **Understanding the Interplay between Memorization and Learning in Large Language Models.**
  [NeurIPS DB 2026] *The Fortieth Annual Conference on Neural Information Processing Systems Evaluations and Datasets Track*
  Bishwamittra Ghosh, Soumi Das, **Qinyuan Wu**, Mohammad Aflah Khan, Krishna P. Gummadi, Evimaria Terzi, Deepak Garg

- **Fine-tuning vs. In-context Learning in Large Language Models: A Formal Language Learning Perspective.**
  [ACL 2026] [Main] [SAC Highlight Award] *The 64th Annual Meeting of the Association for Computational Linguistics*
  Bishwamittra Ghosh, Soumi Das, Till Speicher, **Qinyuan Wu**, Mohammad Aflah Khan, Deepak Garg, Krishna P. Gummadi, Evimaria Terzi

- **Understanding Memorisation in LLMs: Dynamics, Influencing Factors, and Implications.**
  arXiv preprint, 2024.
  Till Speicher, Mohammad Aflah Khan, **Qinyuan Wu**, Vedant Nanda, Soumi Das, Bishwamittra Ghosh, Krishna P. Gummadi, Evimaria Terzi.

2. Memory & Retrieval

From estimating latent factual knowledge to studying deployed memory systems and episodic recall: what information is available, and what can be retrieved?

- ★ **Towards Reliable Latent Knowledge Estimation in LLMs: Zero-Prompt Many-Shot Based Factual Knowledge Extraction.**
  [WSDM 2025] *ACM International Conference on Web Search and Data Mining.*
  Qinyuan Wu, Mohammad Aflah Khan, Soumi Das, Vedant Nanda, Bishwamittra Ghosh, Camila Kolling, Till Speicher, Laurent Bindschaedler, Krishna P. Gummadi, Evimaria Terzi.

- ★ **The Algorithmic Self-Portrait: Deconstructing Memory in ChatGPT.**
  [WWW 2026] [Oral] *The ACM Web Conference*
  Abhisek Dash*, Soumi Das*, Elisabeth Kirsten*, **Qinyuan Wu***, Sai Keerthana Karnam, Krishna P. Gummadi, Thorsten Holz, Muhammad Bilal Zafar and Savvas Zannettou. *(* Equal contribution.)*

– **Temporal Context Reinstatement Drives Episodic-Like Order Memory in Long-Context Language Models.**
[ICML 2026] *The Forty-Third International Conference on Machine Learning*
Mathis Pink, Vy A. Vo, **Qinyuan Wu**, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva

– **Position: Episodic Memory is the Missing Piece for Long-Term LLM Agents.**
arXiv preprint, 2025.
Mathis Pink, **Qinyuan Wu**, Vy Ai Vo, Javier Turek, Jianing Mu, Alexander Huth, Mariya Toneva.

3. Behavior & Agentic Action

From deciding when to call a tool to deciding how to use its output: how should internal knowledge and external evidence jointly guide agent behavior?

– ★ **To Call or Not to Call: A Framework to Assess and Optimize LLM Tool Calling.**
Preprint, under-review
Qinyuan Wu, Soumi Das, Mahsa Amani, Arijit Nag, Seungeon Lee, Krishna P. Gummadi, Abhilasha Ravichander, Muhammad Bilal Zafar

– ★ **When What Agents Know Differs from What Agents Are Told: Analyzing and Optimising LLMs' Deference to Tool Outputs.**
Under-review
Qinyuan Wu, Mohammad Aflah Khan, Sheikh Shafayat, Seungeon Lee, Mahsa Amani, Soumi Das, Krishna P. Gummadi, Manish Gupta, Abhilasha Ravichander

– **In Agents We Trust, but Who Do Agents Trust? Latent Source Preferences Steer LLM Generations.**
[ICLR 2026] *The Fourteenth International Conference on Learning Representations*
Mohammad Aflah Khan, Mahsa Amani, Soumi Das, Bishwamittra Ghosh, **Qinyuan Wu**, Krishna P. Gummadi, Manish Gupta, Abhilasha Ravichander.

– **Characterizing Web Search by Conversational LLM Agents: From Search Decisions and Strategies to Results and Responses.**
Preprint
Mahsa Amani, Seungeon Lee, Abhisek Dash, Asmaa El Fraihi, Yunah Jang, Elisabeth Kirsten, **Qinyuan Wu**, Krishna P. Gummadi, Manish Gupta, Abhilasha Ravichander, and Muhammad Bilal Zafar, Soumi Das

– **Characterizing Web Search in The Age of Generative AI.**
[ACL 2026] [Findings] *The 64th Annual Meeting of the Association for Computational Linguistics*
Elisabeth Kirsten, Jost Große Perdekamp, **Qinyuan Wu**, Mihir Upadhyay, Krishna P. Gummadi, Muhammad Bilal Zafar

Work Experience

ByteDance (TikTok) June 2026 – August 2026
Research Intern: Human Modeling for Personalized LLM Agents.
Developed a leakage-controlled benchmark and hierarchical memory pipeline for predicting future user behavior from longitudinal digital traces. Scaled evaluation and dataset construction to thousands of users, and fine-tuned internal LLMs to improve accuracy by 12% over the base model.

Talks

IASEAI annual conference 2026, UNESCO, Paris, France
Rote Learning Considered Useful: Generalizing over Memorized Data in LLMs
Data Science for Humanity Group 2025, MPI-SP, Bochum, Germany
Rote Learning Considered Useful: Generalizing over Memorized Data in LLMs
AI and Society Group 2024, Ruhr Universität Bochum, Germany
Reliable Knowledge Estimation of LLMs

Awards

SAC Highlight Award, ACL 2026 2026
First-class Scholarship (Top 10%) 2019–2021

Teaching Experience

Selective Topics of Post Training (TA) Winter 2026–27, Saarland University

Efficient Training of Large Language Models (TA) Winter 2025–26, Saarland University
Systems for Large (Language) Models (TA) Winter 2023–24, Saarland University

Mentoring

Nour El Imane EL BAR Oct 2026 – present
Master student at Ecole nationale Superieure d'Informatique (ESI).
Research topic: Personalization of conversational AI agents.

Amal Messaoud Oct 2026 – present
Master student at Ecole nationale Superieure d'Informatique (ESI).
Research topic: Information extraction from social media and reuse for AI agents.

Academic Service

Reviewer: ICLR, ICML, FAccT, COLM, NeurIPS, ARR reviewing cycles, IASEAI

Languages

Chinese (Native), English (Fluent), German (Beginner)

References

Krishna P. Gummadi MPI-SWS, Scientific Director
gummadi@mpi-sws.org
Muhammad Bilal Zafar Ruhr University Bochum, Professor
bilal.zafar@rub.de